

Future Stories (FSCE)-Style Sample Paper — Adventure & Beacon Maths

Time allowed: 45 minutes**Total marks:** 40**Questions:** 40**Calculator:** Not allowed

Name: _____

Date: _____

School: _____

Candidate number: _____

Instructions

Answer on the separate answer sheet using a black pen. Only the answer sheet is marked, not this booklet.

Section 1 (Adventure-style, multiple choice): shade one oval for each question. There is only one correct answer.

Section 2 (Beacon-style, short written answers): write one digit in each box. Give your answer in the format the question asks for. You will get no marks if the format is wrong.

There is space for working out, but you will not get marks for working.

You will not lose marks for a wrong answer, so try every question. If you get stuck, move on and come back to it.

Diagrams are not drawn to scale. You may not use a calculator.

Section 1: Multiple Choice Questions 1 to 28

- 1 What number does the Roman numeral XLVII represent?

[1]

☐ A 47☐ B 57☐ C 43☐ D 52

Working (no marks for working)

- 2 One January morning, four locations record these temperatures: Riverside -6°C , Hilltop -2°C , Valley 3°C , Lakeside -9°C . Which location recorded the coldest temperature?

[1]

MORNING TEMPERATURES

☐ A Lakeside☐ B Riverside☐ C Hilltop☐ D Valley

Working (no marks for working)

- 3 Round 68,472 to the nearest thousand.

[1]

ROUNDING TO THE NEAREST 1,000



- ☐ A 68000 ☐ B 69000
☐ C 68500 ☐ D 68400

Working (no marks for working)

- 4 In the number 803,516, what is the value of the digit 5?

[1]

PLACE VALUE OF 803,516

100,000s	10,000s	1,000s	100s	10s	1s
8	0	3	5	1	6

- ☐ A 500 ☐ B 5000
☐ C 50 ☐ D 5

Working (no marks for working)

- 5 At 6am the temperature was -4°C . By midday it had risen by 11°C . What was the temperature at midday? [1]

- ☐ A 7°C ☐ B 15°C
☐ C -7°C ☐ D 6°C

Working (no marks for working)

- 6 A school gate has the founding year carved into the stonework as MCMXLVIII. In which year was the school founded? [1]

- ☐ A 1948 ☐ B 1947
☐ C 1958 ☐ D 1848

Working (no marks for working)

- 7 Work out $8 + 6 \times (9 - 4)$.

[1]

- ☐ A 38 ☐ B 70
☐ C 34 ☐ D 42

Working (no marks for working)

- 8 Which of these numbers is a factor of 84?

[1]

- ☐ A 14 ☐ B 9
☐ C 16 ☐ D 25

Working (no marks for working)

9 Which of these numbers is a prime number?

[1]

A 61

B 51

C 57

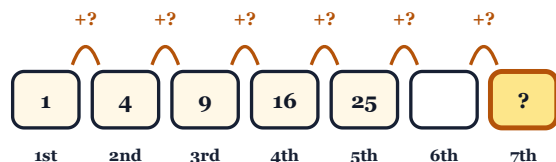
D 63

Working (no marks for working)

10 The sequence of square numbers begins 1, 4, 9, 16, 25, ... What is the 7th square number?

[1]

SQUARE NUMBERS



A 49

B 36

C 42

D 64

Working (no marks for working)

11 What is the value of 5^3 (5 cubed)?

[1]

A 125

B 15

C 25

D 10

Working (no marks for working)

12 Which fraction is equivalent to $\frac{3}{4}$?

[1]

THE FRACTION $\frac{3}{4}$



4 equal parts

A $\frac{9}{12}$

B $\frac{6}{9}$

C $\frac{8}{10}$

D $\frac{5}{8}$

Working (no marks for working)

13 Convert $\frac{7}{20}$ to a decimal.

[1]

A 0.35

B 0.7

C 0.135

D 3.5

Working (no marks for working)

14 Which of these is the smallest: $\frac{3}{8}$, $\frac{2}{5}$, $\frac{1}{3}$, or 0.3? [1]

☐ A 0.3

☐ B $\frac{3}{8}$

☐ C $\frac{2}{5}$

☐ D $\frac{1}{3}$

Working (no marks for working)

15 Work out $\frac{1}{4} + \frac{1}{6}$, giving your answer in its simplest form. [1]

☐ A $\frac{5}{12}$

☐ B $\frac{2}{10}$

☐ C $\frac{1}{5}$

☐ D $\frac{7}{12}$

Working (no marks for working)

16 Round 4.267 to 1 decimal place. [1]

☐ A 4.3

☐ B 4.2

☐ C 4.27

☐ D 4.1

Working (no marks for working)

17 Simplify the ratio 18 : 24 to its simplest form. [1]

☐ A 3 : 4

☐ B 6 : 8

☐ C 9 : 12

☐ D 2 : 3

Working (no marks for working)

18 A pancake recipe for 4 people uses 200 g of flour. How much flour is needed to make pancakes for 10 people? [1]

☐ A 500 g

☐ B 450 g

☐ C 400 g

☐ D 550 g

Working (no marks for working)

19 In a school choir, the ratio of sopranos to altos is 3 : 2, and the ratio of altos to basses is 4 : 3. If there are 8 altos, how many basses are there? [1]

☐ A 6

☐ B 8

☐ C 4

☐ D 10

Working (no marks for working)

20 Three angles lie on a straight line. Two of them measure 55° and 68° . What is the size of the third angle? [1]

☐ A 57°

☐ B 65°

☐ C 62°

☐ D 50°

Working (no marks for working)

21 A triangle has one angle of 90° and another of 34° . What is the size of the third angle? [1]

☐ A 56°

☐ B 46°

☐ C 50°

☐ D 60°

Working (no marks for working)

22 Which of these is NOT a property of a cube?

[1]

- ☐ A It has 5 faces
☐ C It has 8 vertices

- ☐ B It has 6 faces
☐ D It has 12 edges

Working (no marks for working)

23 How many lines of symmetry does a regular hexagon have?

[1]

- ☐ A 6
☐ C 4

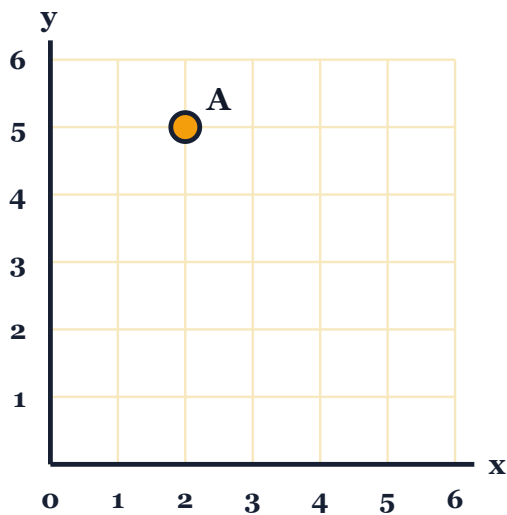
- ☐ B 5
☐ D 3

Working (no marks for working)

24 Point P is plotted on the grid at (2, 5). Point Q is 3 units to the right and 2 units down from P. What are the coordinates of Q?

[1]

POINT P

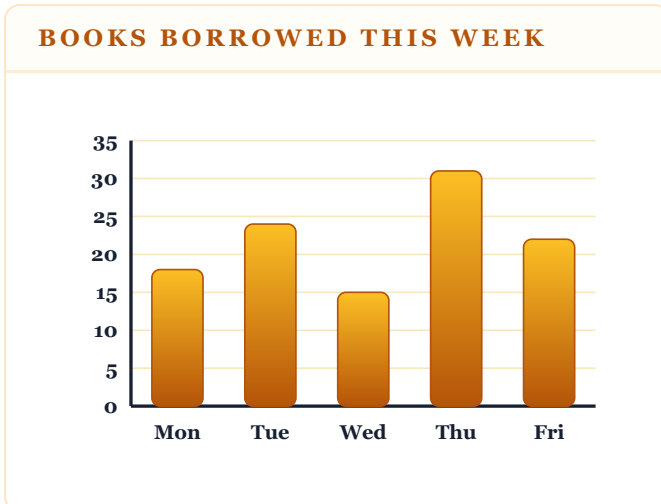


- ☐ A (5, 3)
☐ C (2, 3)

- ☐ B (5, 7)
☐ D (5, 2)

Working (no marks for working)

- 25 The bar chart shows the number of books borrowed from the school library each day one week. How many more books were borrowed on Thursday than on Monday? [1]



- ☐ A 13 ☐ B 12
☐ C 9 ☐ D 16

Working (no marks for working)

- 26 The table shows how pupils in Year 5 travel to school. How many more pupils walk than travel by bus? [1]

HOW YEAR 5 TRAVEL TO SCHOOL

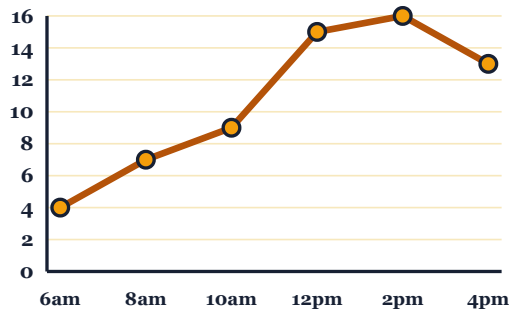
Method	Number of pupils
Walk	34
Car	21
Bus	19
Bicycle	16

- ☐ A 15 ☐ B 13
☐ C 18 ☐ D 21

Working (no marks for working)

- 27 The line graph shows the temperature recorded at a weather station every 2 hours during one day. Between which two times did the temperature rise the most? [1]

WEATHER STATION TEMPERATURE

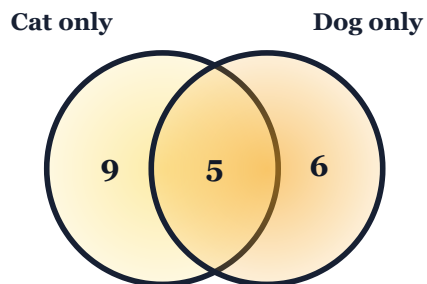


- ☐ A 10am and 12pm ☐ B 6am and 8am
☐ C 12pm and 2pm ☐ D 2pm and 4pm

Working (no marks for working)

- 28 In a class of 30 pupils, 9 have a pet cat only, 6 have a pet dog only, and 5 have both a cat and a dog. How many pupils have neither a cat nor a dog? [1]

PETS IN THE CLASS



- ☐ A 10 ☐ B 8
☐ C 11 ☐ D 14

Working (no marks for working)

Section 2: Short Written Answers Questions 1 to 12

- 1 Work out 236×34 .

[1]

Write one digit in each box.

Working (no marks for working)

- 2 A youth club has 187 pupils to transport in minibuses that each hold 12 pupils. What is the smallest number of minibuses needed so that every pupil has a seat?

[1]

Write one digit in each box.

Working (no marks for working)

- 3 Three classes collected paper for recycling over a week, as shown in the table. The school's target was 400 kg. By how many kilograms did the three classes miss the target?

[1]

PAPER COLLECTED THIS WEEK

Class	Paper collected
Class A	128 kg
Class B	96 kg
Class C	154 kg

 kg

Write one digit in each box.

Working (no marks for working)

- 4 A charity fun run is 15 km long. A runner has completed $\frac{2}{5}$ of the distance. How many kilometres has she run?

[1]

THE FRACTION $\frac{2}{5}$



5 equal parts

 km

Write one digit in each box.

Working (no marks for working)

- 5 A stadium has 25% of its 3,200 seats reserved for members. How many seats are reserved for members? [1]

Write one digit in each box.

Working (no marks for working)

- 6 A bike originally costs £120. In a sale it is reduced by 15%. What is the sale price? [1]

£

Write one digit in each box.

Working (no marks for working)

- 7 Joe and Priya share £360 in the ratio 4 : 5. How much does Priya, who has the larger share, receive? [1]

JOE : PRIYA

Joe

Priya

£

Write one digit in each box.

Working (no marks for working)

- 8 A map has a scale of 1 cm : 5 km. The distance between two villages on the map is 6.4 cm. What is the real distance between the villages? [1]

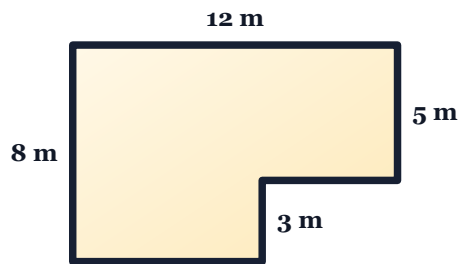
 km

Write one digit in each box.

Working (no marks for working)

- 9 The diagram shows an L-shaped garden. The outer width is 12 m and outer height is 8 m, with a 5 m by 3 m rectangle cut from the bottom-right corner. What is the perimeter of the garden? [1]

L-SHAPED GARDEN



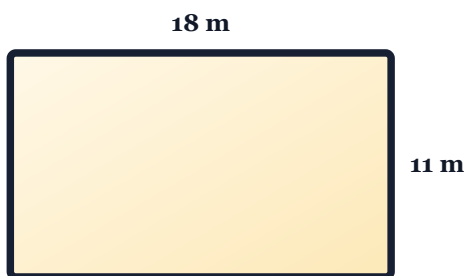
m

Write one digit in each box.

Working (no marks for working)

- 10 A rectangular sports hall floor measures 18 m by 11 m. What is its area? [1]

SPORTS HALL FLOOR



m²

Write one digit in each box.

Working (no marks for working)

- 11 Five friends timed how long it took them to solve a puzzle: 8 minutes, 11 minutes, 6 minutes, 13 minutes and 12 minutes. What was their mean (average) time? [1]

minutes

Write one digit in each box.

Working (no marks for working)

- 12 A school trip coach leaves at 09:45 and arrives 2 hours and 50 minutes later. At what time does it arrive? [1]

DEPARTURE TIME



:

Answer using the 24-hour clock (HH:MM).

Working (no marks for working)

This was the last question. You can use any spare time to check your work.

Answer Sheet — Maths

Use: black pen

Name:

Date:

School:

Candidate number:

Section 1: Multiple choice. Shade one oval per question.

1 A ☐ B ☐ C ☐ D ☐	2 A ☐ B ☐ C ☐ D ☐	3 A ☐ B ☐ C ☐ D ☐	4 A ☐ B ☐ C ☐ D ☐	5 A ☐ B ☐ C ☐ D ☐	6 A ☐ B ☐ C ☐ D ☐
7 A ☐ B ☐ C ☐ D ☐	8 A ☐ B ☐ C ☐ D ☐	9 A ☐ B ☐ C ☐ D ☐	10 A ☐ B ☐ C ☐ D ☐	11 A ☐ B ☐ C ☐ D ☐	12 A ☐ B ☐ C ☐ D ☐
13 A ☐ B ☐ C ☐ D ☐	14 A ☐ B ☐ C ☐ D ☐	15 A ☐ B ☐ C ☐ D ☐	16 A ☐ B ☐ C ☐ D ☐	17 A ☐ B ☐ C ☐ D ☐	18 A ☐ B ☐ C ☐ D ☐
19 A ☐ B ☐ C ☐ D ☐	20 A ☐ B ☐ C ☐ D ☐	21 A ☐ B ☐ C ☐ D ☐	22 A ☐ B ☐ C ☐ D ☐	23 A ☐ B ☐ C ☐ D ☐	24 A ☐ B ☐ C ☐ D ☐
25 A ☐ B ☐ C ☐ D ☐	26 A ☐ B ☐ C ☐ D ☐	27 A ☐ B ☐ C ☐ D ☐	28 A ☐ B ☐ C ☐ D ☐		

Section 2: Short written answers. One digit per box.

Q1

Q2

Q3 kg

Q4 km

Q5

Q6 £

Q7 £

Q8 km

Q9 m

Q10 m²

Q11 minutes

Q12 :

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